

[498]

On the Inversion of a Quadric Surface.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XI. (1871), pp. 283–288.]

THE inversion to be considered is that by reciprocal radii vectors, viz. if x, y, z are rectangular coordinates, and $r^2 = x^2 + y^2 + z^2$, then x, y, z are to be changed into $\frac{x}{r^2}, \frac{y}{r^2}, \frac{z}{r^2}$. But it is convenient to introduce the homogeneity a fourth coordinate $w, = 1$; and the change then is x, y, z into $\frac{xw}{r^2}, \frac{yw}{r^2}, \frac{zw}{r^2}$.

Starting from the quadric surface

$$(a, b, c, d, f, g, h, l, m, n, x, y, z, w)^2 = 0,$$

or, what is the same thing,

$$\begin{aligned} & (a, b, c, f, g, h, l, m, n, x, y, z)^2 \\ & + 2w(lx + my + nz) \\ & + dw^2 = 0, \end{aligned}$$

the equation of the inverse surface is

$$\begin{aligned} & w^2(a, b, c, f, g, h, l, m, n, x, y, z)^2 \\ & + 2w(lx + my + nz)r^2 \\ & + dr^4 = 0, \end{aligned}$$

where $r^2 = x^2 + y^2 + z^2$; the inverse surface is then a quartic having the nodal conic $w = 0$, $x^2 + y^2 + z^2 = 0$ (circle at infinity); and having the node $x = 0, y = 0, z = 0$ (the origin of inversion); or say it is a nodal binodular quartic surface, or nodal anallagmatic quartic.

For x, y, z write $a = \frac{1}{4}ar, y = \frac{1}{4}\beta r, z = \frac{1}{4}\gamma r$, and put for shortness

$$\begin{aligned} l\alpha + m\beta + n\gamma &= A, \\ al + bm + cn &= a', \\ bl + cm + an &= b', \\ cl + am + bn &= c', \end{aligned} \quad (a, b, c, f, g, h, l, m, n)^2 = J,$$

then

$$r^2 \text{ becomes } r^2 = \frac{w^2}{r^2} + \frac{1}{4} \frac{w}{r}.$$

$$lx + my + nz = \frac{1}{4} \frac{w}{r} A,$$

$$(a, b, \dots, x, y, z)^2 = (a, b, \dots, \alpha, \beta, \gamma)^2 w^2 - (ax + by + cz) \frac{w}{r} + \frac{1}{4} J \frac{w^2}{r^2}.$$

Hence the equation is

$$\begin{aligned} & d \left\{ r^2 - 2r \frac{w}{r} + w^2 \left(\frac{1}{4} \frac{a^2}{r^2} \right) \right\} - \frac{2mwr}{r} + \frac{w^2}{r^2} \left(\frac{1}{4} \frac{A}{r} \right) \\ & + 2 \left\{ aw - \frac{w^2 r}{w} + \frac{1}{4} A \frac{w^2}{r^2} \right\} \left(w + \frac{1}{4} \frac{w}{r} \right) \\ & + w^2 [(a, \dots, \alpha, \beta, \gamma)^2 - (ax + by + cz)] = 0; \end{aligned}$$

viz. arranging and reducing, this is

$$\frac{dr^2}{w^2} + w^2 \left\{ -\frac{1}{4} \frac{A}{r^2} r - \frac{w^2 r}{r} - (a, \dots x, y, z)^2 \right\} + w \left\{ \frac{2a}{r} - \frac{1}{4} (ax + by + cz) \right\} + w^2 \left\{ -\frac{1}{2} \frac{2}{r^2} + \frac{1}{4} A \frac{w^2}{r^2} \right\} = 0;$$

and we may without loss of generality assume

$$\begin{aligned} -\frac{w^2}{r^2} + f - 0, \quad \text{this is } d f - mn = 0, \\ -\frac{a^2}{r^2} + g - 0, \quad e - mn = 0, \\ \text{etc.} \end{aligned}$$

The equation then is

$$\begin{aligned} & \frac{r^4}{w^4} + w^2 \left\{ -\frac{1}{4} \frac{a^2}{r^2} (x^2 + y^2 + z^2) + \left(\frac{1}{4} \frac{f^2}{w} \right) y^2 + \dots \right\} \\ & + w^2 \left\{ \frac{2a}{r} - \frac{1}{4} (ax + by + cz)r \right\} \\ & + w^2 \left\{ -\frac{1}{2} \frac{2}{r^2} + \frac{1}{4} A \frac{w^2}{r^2} \right\}. \end{aligned}$$

Write

$$ad - D = a'd, \quad bd - m' = b'd, \quad cd - n' = c'd.$$

We have

$$a = al + bm + pn = pa = a'l + \frac{bm^2}{d} + \frac{ln^2}{d} = \frac{1}{d}(ad - D + b),$$

that is

$$a = b'l + \frac{la}{d},$$

and similarly

$$b = a'l + \frac{ma}{d}, \quad c = a'l + \frac{na}{d}.$$

Hence also

$$A = la' + mb' + nc' = \frac{a'}{d}.$$

and the equation is

$$\begin{aligned} & \frac{aw^4}{r^4} + w^2 \left\{ -\frac{1}{4} \frac{a^2}{r^2} x^2 + c'^2 + \left(\frac{1}{4} \frac{f^2}{w} \right) y^2 + \dots \right\} \\ & + w^2 \left\{ -\frac{b'}{r} cx + \frac{c'^2}{r^2} cx - \frac{m^2 r^2}{r^2} \right\} \\ & + w^2 \left\{ \frac{1}{4d} (b'^2 + n'^2) xw + n'(cr^2 + wc^2) \right\}. \end{aligned}$$

This is Kummer's form, viz.

$$r^4 = ka^2(n^2x^2 + \beta y^2 + \gamma z^2) + lawr + 2w(ax + by + cz)r,$$

where

$$\begin{aligned} -4a, +k^2 &= -\frac{1}{4}\frac{a^2}{r^2}, \\ -k\beta, + &= -\frac{1}{4}\frac{b^2}{r^2}, \\ -k\gamma, + &= -\frac{1}{4}\frac{c^2}{r^2}, \\ -k\lambda, &= \frac{fe}{da}(ba' + ab'c) + \beta\frac{m}{d}. \end{aligned}$$

Hence Kummer's equation

$$\beta_4 = k\lambda = \frac{a^2}{\lambda + a} = \frac{b^2}{\lambda + \beta} = \frac{c^2}{\lambda + \gamma},$$

or say

$$6k\lambda + 6k\lambda^2 = \frac{216a^2}{4\lambda + 4a} \cdot \frac{4aa^2}{4\lambda + 4\beta} \cdot \frac{216c^2}{4\lambda + 4\gamma},$$

becomes

$$6k\lambda^2 = \frac{4}{d^2}(b'^2a + a'b'^2 + c'^2) - \frac{a'^2}{a^2\left(\frac{a^2}{d^2} - \frac{a^2}{d^2} + 4a\right)} - \frac{4ab'^2}{a^2\left(\frac{b^2}{d^2} - \frac{b^2}{d^2} + 4b\right)} - \frac{4ac'^2}{a^2\left(\frac{c^2}{d^2} - \frac{c^2}{d^2} + 4c\right)},$$

which is satisfied by $4\lambda = -\frac{1}{4}\frac{a^2}{d}$. Writing therefore

$$6\lambda + \frac{1}{4}\frac{a^2}{d} = -\frac{d}{a},$$

that is

$$\begin{aligned} \delta\lambda &= -\frac{2D}{a} - \frac{a^2}{d^2}, \\ 6a\lambda^2 &= \frac{4D^2}{d^2} + \frac{4Da}{a^2} + \frac{a^4}{d^4}; \end{aligned}$$

the equation is

$$\frac{d}{r^4} = \frac{d}{w^4} - \frac{4}{d^2}b'a' + a'b'^2 + c'^2 = \frac{4aa^2}{d^2\left(\frac{a^2}{d^2} - \frac{a^2}{d^2} + 4a\right)} - \frac{4ab'^2}{d^2\left(\frac{b^2}{d^2} - \frac{b^2}{d^2} + 4b\right)} - \frac{4ac'^2}{d^2\left(\frac{c^2}{d^2} - \frac{c^2}{d^2} + 4c\right)},$$

viz. this is

$$Da^2 + aB - Da = Da^2 - \frac{2D^2}{d} - \frac{aa'a^2}{d^2\left(\frac{a^2}{d^2} - \frac{a^2}{d^2}\right) + 4a} + \dots$$

which is of course satisfied by $\theta = 0$. Moreover the derived equation

$$= a - 2D\delta = \frac{Da^2}{d(b+d)^2a^2c'^2\left(\frac{a^2}{d}\right)^2} + \frac{aa^2c^2}{d(b+d)^2} + \frac{aa'a^2}{d(b+d)^2}$$

is also satisfied by $\theta = 0$, so that this is a double root. The equation in fact is

$$(P_a^2 + Pa - Da)(a^2 + a^2c'^2(B+a)(d+b)(d+c)) \\ + (Pa + db'^2(B+c)(d+a)(d+c) - 3a'^2c^2(B+a)(B+b)(d+c)) = 0,$$

or, expanding and dividing by θ^2 , this is

$$(Pa^2 + a)(d+a)(d+b)(d+c) + 2a'b'(d+a)(d+b)(d+c)c \\ - (Pb'^2 + ac'(b+c)c)(a' + c'(d+a)(d+b)(d+c)) = 0.$$

[499]

On the Theory of the Curve and Torse.*[From the Quarterly Journal of Pure and Applied Mathematics, vol. XI. (1871), pp. 294–317.]*

THE fundamental relations in the theory of the Curve and Torse were first established in my “Mémoire sur les Courbes à double courbure et sur les Surfaces développables,” *Liouv. t. x.* (1845); [30] see also *Cambr. and Dubl. Math. Jour., t. iv.* (1850), [85], viz. I showed that the systems (m, r, β, h, x, y) and $(r, n, \alpha, g, y, \beta)$, the notation is anallagmatically explained, each of them satisfied the Plückerian relations. An additional set of equations giving the values of g, β, h, C, g', h' was furnished by Dr Salmon’s “Theory of Reciprocal Surfaces.” *Trans. R. I. Acad., t. xxiii.* (1847); see also his *Solid Geometry*. The theory as thus established is complete in itself, but the account of certain singularities α, H, G ; the singularity α was first considered in my paper “On a special sextic Developable,” *Quart. Math. Jour., t. vii.* (1865), [372], (there called θ), and I afterwards endeavoured to take account of the remaining singularities H, G . I was in correspondence on the subject with Prof. Cremona, and the discovery of the complete forms of several of the formulæ is due to him.

There has recently appeared a very valuable memoir by M. Zeuthen, “Sur les singularités ordinaires d’une courbe gauche et d’une surface développable,” *Annali di Matem. t. iii.* (1869), in which, however, from considerations the singularity α and does not throughout attend to H, G .

I propose in the present memoir to reproduce and develop the whole theory.

Explanations and Notation.

1. We have a singly infinite series of points, lines, and planes; viz. each line passes through two consecutive points and lies in two consecutive planes; each plane passes through three consecutive points and contains two consecutive lines; each point

is the intersection of three consecutive planes and of two consecutive lines. The points describe and the lines envelope a *curve*; the lines describe and the planes envelope a *torse*; the entire system of points, lines, and planes is thus the system of the curve and torse. The curve is the edge of regression or cuspidal curve of the torse; in regard to the curve the points are points (or incunabula), the lines tangents, and the planes osculating planes; in regard to the torse the points are points on the edge of regression, the lines generating lines, and the planes tangent planes.

2. Each line of the system is met by a certain number of non-consecutive lines, and the locus of the points of intersection (or say the locus of the intersection of two intersecting non-consecutive lines) is a nodal curve on the torse, or say simply it is the *nodal curve*. The plane containing the two intersecting non-consecutive lines envelopes a torse which is called the *nodal torse*.

3. There is occasion to consider

m , order of the system; this is the number ∞ of points of the system which lie in a given plane; or it is the order of the curve.

r , rank of the system; this is the number of lines of the system which meet a given line. It is thus the class of the curve; and the order of the torse.

n , class of the system; this is the number of planes of the system which pass through a given point. It is thus the class of the torse.

α , number of stationary planes; that is, planes each passing through four consecutive points of the system.

β , number of stationary points, that is, points each of them the intersection of four consecutive planes of the system.

g , number of lines in two planes (that is, lines each of them the intersection of two non-consecutive planes of the system) contained in a given plane; or say, number of apparent double planes of the torse.

G , number of double planes, or *tropes*, of the torse; viz. considering the torse as the envelope of a variable plane, each of these is a position of this plane, equivalent to two; or say the plane in question is a *trope*.

h , number of *lines through two points* (that is, lines each through two non-consecutive points of the system) passing through a given point; or say, number of apparent double points of the curve.

H , number of double lines, or *nodes*, of the curve; viz. considering the curve as the locus of a variable point, if the point in the course of its motion comes twice into the same position, we have then a double point or node.

x , number of *points in two lines* (that is, points each on two non-consecutive lines of the system); or say, number of apparent double points of the curve.

σ , number of stationary lines of the system, that is, lines each containing three consecutive points of the system.

α , number of double lines of the system, that is, lines each containing two pairs of consecutive points of the system.

y , number of points in two lines (that is, points each of them the common intersection of three non-consecutive lines of the system); these are also triple points on the curve.

Y , number of apparent double points of nodal curve.

z , class of nodal curve.

ζ , number of *planes through three lines* (that is, planes each of them through three non-consecutive lines of the system); these are also triple tangent planes of the torse.

y' , number of planes of the system each of them passing through a non-consecutive line of the system; these are also tangent planes of the torse, and nodal torse, stationary planes of the latter torse.

Y' , number of apparent double planes of nodal torse.

z' , order of nodal torse.

x' , class of nodal torse.

4. The formulæ thus contains in all the 23 quantities

$$m, r, n, \alpha, \beta, g, G, h, H, x, y, \sigma, \alpha, Y, z, \zeta, y', Y', z', x'.$$

My own Plückerian equations, or, say, the Plücker–Cayley equations, establish in all 6 relations between the first 12 quantities, and thus enable the expression of them in terms of any seven, say of

$$m, r, n, G, H, x, x.$$

and the Salmon-Cremona equations then lead to the expressions in terms of these, of the remaining eight quantities y, y, b, g', C, x', g' .

I also consider

D_m , the deficiency of the curve,

D_n , the deficiency of the nodal curve.

I will first consider the equations themselves, and the more algebraical transformations thereof, and afterwards the geometrical theory.

The Plücker-Cayley Equations.

5. These are found (as will be further explained) by considering first the cone, vertex an arbitrary point, which passes through the curve; and secondly, the section of the torse by an arbitrary plane. We have in the two figures

Cone.	Section.
m , order,	n , class,
r , class,	r , order,
$h + H$, double lines,	$g + G$, double tangents,
y , double points,	β , stationary tangents,
$x + \alpha$, stationary planes.	$\alpha + x$, stationary points.

And hence the two sets of quantities respectively are connected by the Plückerian relations, viz. these are

$$\begin{aligned}
 r &= m(m-1) - 2(h+H) - 3y, \\
 n + 6m &= 3m(m-1) - 6(h+H) - 8y, \\
 y + a &= \frac{1}{2}m(m-1)(m-2) - 9y, \\
 &\quad - (m-1)(8h+H) - (8-8)y \\
 &\quad - \frac{1}{4}m(m-1)(m-2) - 2(8h+H), \\
 &\quad - (8+2)y, \\
 m &= r(r-1) - 2(g+G) - 3(x+\alpha), \\
 B &= 3r(r-1) - 6(g+G) - 8(x+\alpha), \\
 b + \beta &= \frac{1}{2}r(r-1)(r-2) - 9(x+\alpha), \\
 &\quad - (r-1)(8h+H), \\
 &\quad - (8-8)(x+\alpha).
 \end{aligned}$$

and combining the two systems

$$\begin{aligned}
 \beta &= a + 2(n - r), \\
 y &= a + (n - r), \\
 h + H &= x + y + (m - n)(n + r) \\
 &\quad + \frac{1}{2}(n - 1)(n - 5) - 4(g + G) + B = 2(m - 1)(n - 7) + 6 - \\
 \frac{1}{4}(r - 1)(n - 7) + 6 - 4(g + G) + (m - 1)(n - 2) + (8 - x)(y - y) + (a - 1) - (n + a) \\
 &= 0.
 \end{aligned}$$

6. Taking as data r, m, n, G, H, a, x , we find very easily

$$\begin{aligned}
 h &= \frac{1}{2}m^2 - 10m - 5n + (n - r) - H, \\
 g &= \frac{1}{2}n^2 - 10n - 5m + (m - r) - G, \\
 \alpha &= 3(n - r) - r - n - 5m - 5a - 5x, \\
 \beta &= \frac{1}{2}y(n^2 - r) - n - 5n - 5a - 5x, \\
 a &= n - 2r + 3x + a, \\
 \beta &= a - 2r + 3a + a.
 \end{aligned}$$

The Salmon-Cremona Equations.

7. These are

$$\begin{aligned}
 a(r - 2) &= 2a + 4\beta + g + 8a + 4H, \\
 \sigma(r - 2) &= 4n - 4\beta + g + 8a + 14H, \\
 x(r - 2) &= 4n - 4\beta - 4r + 1B - 14H + (3n - 10) + 12H, \\
 &\quad + a(8r - 16) - 19a - 2a, \\
 y(r - 2) &= (m + n)(r - 4) - 2(n - r) + 4\beta - 3n - (4r - 5)q - (3r - 10) + 12H, \\
 g + Z &= 9r(r - 2) - 8r + 8a + 9b + (4r - 16)a - 14a - 12n, \\
 \zeta + \beta &= 3r(r - 2) - 4r - 4r + 4y(8r - 14) + (3r + 14)\beta + 12H, \\
 \beta' &= z(r - 2)y' - y + 24Y'.
 \end{aligned}$$

where the second values of y, y' respectively are reduced forms obtained by the aid of the foregoing Plückerian relations.

8. Expressing these in terms of r, m, n, G, H, α, x , we obtain

$$\begin{aligned}
 y &= \frac{1}{4}m + 12r - 14m - 6x + 8x + 4H, \\
 b &= \frac{1}{2}(r - 5r - 10r) + 5x(n + 8r + 18r) + 42x + 76r + 24x, \\
 &\quad - \frac{1}{4}(r - 8r + 11r - 60(r + 25r - 25) + 8(8r + 14 - 5r + 12n + 24G \\
 &\quad + (n + 5r + 4r + 6) - 28a - 14r + 240n - 120n - 12x + 6x + 24G).
 \end{aligned}$$

$$\begin{aligned}
 y' &= b - 2n - 24r, \\
 a' &= a + 12r - 14m - 6x + 8a + 4G, \\
 b' &= \frac{1}{2}(r - 5r - 10x) - 50(n - 8r - 5x) - 42x + 75n + 24G, \\
 &= \frac{1}{4}(r - 8r + 11x + 60n - 25(c - 7r - 25) - 6(7n - 14)r + 12r - 12x + 24G \\
 &\quad + 4(c + 5r + 14)r - 25a + 14H + 24Gn - 120n - 12x + 6x + 24G). \\
 y' &= b - n + 2n - 24G.
 \end{aligned}$$

9. We have therein

$$\begin{aligned}
 r' &= y - n - 8(n - r) - 4(7 - H), \\
 g' &= y' + 6(m - r), \\
 y' &= y - (n - r)(n + 2r - 5) + 2(g - H), \\
 y &= -\frac{1}{2}(n - r)(r - r) + (8 - r)(r - r) + (g - r) - 2(g - H).
 \end{aligned}$$

10. Instead of obtaining the above values of y, β, g directly it is convenient to verify them by substitution in the equations from which they were obtained; viz. writing for shortness $a+8(n-r)$ for b , these may be written

$$\begin{aligned}
 (n - r - 2) + 8a + 4H + 8a - 6x + 4H &= 0, \\
 -2a(r - 2) - 8r + 18(r - r) + 2y - 4 - 2(M)r + 2 &= 0, \\
 -2r(r - 2) + 8r - 2r - r + 4y(r - 4) - 12(8r - 4)x + 4(8 - 14)H - 4x + 11(n - r) - 24H &= 0, \\
 -2r(r - 2) - 4y - 11(8 - 14)H + (5r - 9 - 24H) &= 0, \\
 -8r + 16r + 18a(r - 6) - (3r - 4(8 - 18r - 12r + 9r - 25 + 24H) &= 0, \\
 -4r(r - 2) - 4y - (r - 6)(r - 4) + 4y(8 - 14)H + 12H + 24H &= 0,
 \end{aligned}$$

which are to be satisfied by

$$\begin{aligned}
 y &= a + 12r - 14a - 6r + 8a + 4H, \\
 \beta &= a + n - 2r + 8a + a + a + 4H, \\
 \beta &= a + n - 2r + 8a + 4G, \\
 g &= b - n + 2n - 24G, \\
 H &= a + (g - H), \\
 b' &= b + a + b + 8(n - r).
 \end{aligned}$$

11. We have, in fact, then

$$\begin{aligned}
 &-x + r + 2a \\
 &\quad + 2x \\
 &\quad + 12n + 4n - 12r + 4n \\
 &\quad - 6x \\
 &\quad + 8x \\
 &\quad + 4H \\
 &\quad + ar - 14n + 4n + 12r - 8n - 4n + 4H \\
 &\hspace{15em} = 0.
 \end{aligned}$$

secondly

$$\begin{aligned}
 & -r^2 + r^2 \\
 & + 2r^2 - 2x \\
 & + 2r^2 - 2P \\
 & - 8r \\
 & - 12 \\
 & + 4r + 12n + 4n \\
 & - 2n \\
 & - 14H \\
 & + 7y \\
 & + 6n + 8H - 14n - 4n - 12r + 4H \\
 & + 4 \\
 & + 24H
 \end{aligned} = 0,$$

that is

$$-r^2 + 3r^2 - 10r - 2r^2 = -2P^2 + 2y + 6n + 6n = 0.$$

thirdly

$$\begin{aligned}
 & -r^2 + 5r^2 - 4r \\
 & + r^2 - 5r^2 + 4r + 2P^2(8r - 4r + 11(7-)) \\
 & - 4r + 16r \\
 & + 6n - 18n - 12n \\
 & - 14n + 12n \\
 & + 8r \\
 & - 14n - 5n - 6n + 50n \\
 & + 12n \\
 & - 10n + 16n \\
 & - 14n \\
 & + 9(7 - 8n - 8r) - 8(2H) + 25H \\
 & + r^2 - 9r^2 + 11r + 60n - 8H^2 - 12n - 25H + (-r^2 - 9r + 11r + 60n - 14)\beta - 36r + 8 = 0.
 \end{aligned}$$

$\therefore -r^2 + 5r^2 - 4r$ is here verified, where:

$$\begin{aligned}
 & + r^2 - 6n + 12n^2 + 11r + 60n - F'(-r^2 - r + 16r) + r^2 - 18 + 70n^2 - 16n^2 - 25H \\
 & - 14H + (-r^2 - 9r + 11r + 60n - 14)\beta - 36r + 8H = 0
 \end{aligned}$$

and lastly

$$\begin{aligned}
 & -24P \\
 & -4mr \\
 & +4m+12n+36n+12n \\
 & +8G \\
 & +r^2-5r^2+11r+60n-F'(7m-9m+12n) \\
 & +12n-14n-14n-12n-24n \qquad \qquad \qquad =0. \\
 & -r^4-6r^2-11r+60n-F(7-4)x \\
 & -144x \\
 & +r^4-6r+18n+4n+12n \\
 & +12n+24n+24G \\
 & -8(2r-16r)+8(2r-16)+8r+24H \\
 & \qquad -4r+14r+12-8r+8(r-24)H \\
 & \qquad + (2P-6-6r-6n)x \\
 & \qquad +12n-16x \\
 & \qquad -80n \\
 & \qquad +80n+8r+4n+24G=0,
 \end{aligned}$$

that is

$$\begin{aligned}
 & 2\beta-6r+2x-6r+4(r-24)H-18n+14H+24G \\
 & +r^2-8r+8n+12n+14)\beta-12n-16x-36n+24G \\
 & +r^2-9r+14n-14n+22-4n+4n+24G=0,
 \end{aligned}$$

which completes the verification.

12. The deficiency of the curve is given by the equation

$$2D_m = r - 2m + 2 + \beta,$$

or substituting for β its value $= n - 2r + 8x + \alpha$, this is

$$2D_m = n + n - 2r + 2 + \alpha + x.$$

The deficiency of the nodal curve z is given by

$$2D_z = z - 2x + 2x = (h - r)(g - 1) - 2(g + 8H).$$

which, substituting for z , x , and y , becomes

$$\begin{aligned} e + 8x &= 8r - 8n + 8r - 24a - 12x = 8x = G - 14, \\ \frac{1}{4}(r - 8x + 11x - 60n - F(7m - 9n + 12n)) &= 8H + 12, \\ &= r(8 - 1)D_n = (r - 1)(g - 1) - 2g - 8H. \end{aligned}$$

respectively, become

$$2D_x = r - 14a; 2D_x = a + a + r(z + 8x) + r(8H) + r(z + 8H) - 8\beta - 14H,$$

whence, also, writing herein $a + a + a + 2D + a - 2r$, we have

$$2D_a = (m - 16)D_a = (r - 1)(r - 5) - 8a - 2y' - 12H.$$

a relation between the two deficiencies.

Geometrical Theory of the foregoing Relations.

13. In considering the geometrical theory, we have to speak of the original curve, or curve of the system, and also of the nodal curve; it will be convenient to call them the curve m and the curve n respectively. I speak of the torse absolutely, to signify the torse of the system, as in what follows there is not the inferior expression to make of the nodal torse. I speak also of a torse, meaning thereby any one of the stationary planes, the number of which is $= n$; and so a line, meaning the line that with the curve m , or, in like proof, that a plane, meaning thereby an osculating plane of the curve m ; with the indication of the stationary planes belong to the torse; thus we have a stationary tangents equal to β , a tangent plane equal to $n - 2r + 8x + a$ for the planes belonging to the torse.

In particular the points are now the points of the system which lie in a given plane; the lines n are the lines which meet a given line; the planes α are the planes which pass through the other line.

14. Observe that the expressions, the geometrical meaning of which is at one significant, there are some such expressions which have only a relative significance, viz. in regard to the system considered in connexion with a given point, line, or plane, as the case may require. Thus the expression, the line g , must be understood of the system in connexion with a given plane, viz. signify, the plane g must be understood, the plane g would of course mean the planes of points of the system belonging to the line g .

In particular the points α are the points which lie in a given plane, the lines α are the lines which meet a given line; the planes α are the planes

which pass through a given point. There are two such forms, viz. the radial $\sqrt{p^2} \, rp^2$ is may be either α or α , hence the same is $= 0$.

15. But it is to be further shown that the line α is torsoidal, each sheet of the surface being touched along the line by the plane $\alpha = 0$; we have to show that the

16. The mode of obtaining these equations has already been indicated. We in fact consider the system in connexion with an arbitrary point, and with an arbitrary plane. The point is the vertex of a cone passing through the curve m , and this cone is of the order m , the class r , with $h + H$ double lines, β stationary lines, $y + a$ double tangent planes, and $x + \alpha$ stationary planes; viz. the order of the cone is equal to the number of lines in which this is intersected by a plane, the tangent planes; and so on. Each of these is in fact, for any point of the line we have σ or three points of the curve m , equal to the number of the cone $= m$, equal to the number of stationary planes from the point to it. Hence considering the cone, $y + a$ is the number of apparent double planes, equal to the number of double points A, B, C shall each of them a square; but this being so, the element at D will not be in general a square, and it is only for certain surfaces that it is so. But if (whatever the curves

[500]

On a Theorem relating to Eight Points on a Conic.*[From the Quarterly Journal of Pure and Applied Mathematics, vol. XI. (1871), pp. 344-346.]*

THE following is a known theorem:

“In any octagon inscribed in a conic, the two sets of alternate sides intersect in the 8 points of the octagon and in 8 other points lying in a conic.”

In fact the two sets of sides are each of them a quartic curve, hence any quartic curve through 13 of the $8 + 8$ points passes through the remaining 8 points; but the original conic together with a conic through 8 of the 8 new points form together such a quartic curve; and hence the remaining 5 of the new points (inasmuch as obviously they are not situate on the original conic) must be situate on the conic through the 5 new points; that is the 8 new points must lie on a conic.

We may without loss of generality take (x, y, z) , (x_1, y_1, z_1) , ..., (x_8, y_8, z_8) , as the coordinates (x, y, z) of the 8 points of the octagon, and obtain hereby an *θ* posteriori verification of the theorem, by finding the equation of the conic through the 8 new points: the result contains cyclical expressions of an interesting form.

Calling the points of the octagon 1, 2, 3, 4, 5, 6, 7, 8, the 8 new points are

$$12.45, 23.56, 34.67, 45.78, 56.18, 67.23, 78.34, 81.45.$$

viz. 12.45 is the intersection of the lines 12 and 45; and so on. The 8 points lie on a conic, the equation of which is to be found.

The equation of the line 12 is

$$x : (a_1 + a_2)y + a_1 a_2 z = 0,$$

or as it is convenient to write it

$$x - (1 + 2)y + 12.z = 0,$$

viz. 1, 2, &c., are for shortness written in place of a_1 , a_2 , &c. respectively.

The coordinates of the 12.45 are consequently proportional to the terms of

$$\begin{aligned} 1, - (1 + 2), 12, \\ 1, - (4 + 5), 45, \end{aligned}$$

or say they are as

$$12(4 + 5) - 45(4 + 2) : 12 - 45 : -1 + 2 - 4 + 5.$$

The equation of the line (12.45)(23.56) which joins the points 12.45 and 23.56 then is

$$\begin{vmatrix} x, & y, & z \\ 12(4 + 5) - 45(1 + 2), & 12 - 45, & -1 + 2 - 4 + 5 \\ 23(5 + 6) - 56(2 + 3), & 23 - 56, & -2 + 3 - 5 + 6 \end{vmatrix} = 0,$$

where the determinant vanishes identically if $2 = 5$ or $1 - 8 = 4 - 6$; in fact thereby becomes

$$\begin{vmatrix} x, & y, & z \\ 2'(4 + 5) - 4(2 + 5), & 4(2 - 5), & 4'(1 - 5) \\ 3'(5 + 6) - 5(2 + 6), & 5(3 - 6), & 5'(2 - 6) \end{vmatrix} = 0,$$

which is $= 0$; the determinant divides therefore by $2 - 5$; the coefficient of x is easily found to be

$$= (2 - 5)\{12(4 + 5) - 45(1 + 2) + 23(5 + 6) - 56(2 + 3)\}$$

and so for the other terms; and omitting the factor $2 - 5$ the equation is

$$\begin{aligned} & x\{12 - 23 + 84 - 45 + 56 - 1 + 2 - 5 + 6\} \\ & - y\{12(4 + 5) - 45(1 + 2) + 23(5 + 6) - 56(2 + 3)\} \\ & + z\{-(1 + 2)(2 + 8) - (2 + 8)(5 + 6) + (5 + 6) - (4 + 5)\} = 0. \end{aligned}$$

There is now not much difficulty in forming the equation of the required conic; viz. this is

$$\begin{aligned} & (2 - 5)\{x(2 + 7)y + Cx\} \\ & \times (-8\{12(4 + 5) - 45(2 + 2)\} \\ & + (-3)\{12(4 - 5) - 25(8 - 2)\} \\ & + (-3)\{12(2 - 8) - 25(1 - 2)\} \\ & + (-3)\{56(4 - 5) - 12(1 - 2)\}) \\ & = 0. \end{aligned}$$

[501]

Review. Tables de Logarithmes vulgaires à dix décimales construites d'après un nouveau mode par S. Pineto.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. XI. (1871), pp. 375–376.]

TEN tables containing 36 pages—the principal one being a table in 44 pages, the 22 left-hand pages containing the 10 figure logarithms of the 10000 numbers from 1,000,000 to 1,010,000, and the 22 right-hand pages the proportional parts. 01, 02, . . . 09 of the differences. A like table 100,000 to 999,999 would occupy 2000 pages. By means of an auxiliary table of 5 pages, and of a slight increase of the necessary calculations, the table of 44 pages does the work for the 10 figure logarithms of all numbers up to 1,000,000. This is the principal feature, and (so far as I am aware) of a line, meaning the first eight are between 1000 and 09474 (being the limits of the first five digits of the logarithms in the principal table), the auxiliary table by means of the third figures gives 30, adding $\log \frac{1}{2}$ to B , we have a logarithm included in the limits of

the principal table, and seeking for the corresponding number, this is $= \frac{N}{B}$ number having B for its logarithm; that is, the required number is $= M$ times the number obtained as above. Of course as regards the principal Table, the proportional parts are employed in the usual manner; the tabulation of them to hundredths (instead of tenths) facilitates the interpolation; for better securing the accuracy of the last figure, directions are given in regard to the 11th and 12th figures. An example of the determination of a logarithm is as follows:

$$\begin{aligned}\pi &= .314159\,26536 \\ M &= .32 \\ N\pi &= .100526\,48152 \\ \log \frac{1}{M} &= +.49485\,00216.88 - 10 \\ \log &= +.20020\,26579.73 \\ 64 & \quad 2764.80 \\ 91 & \quad 89.21 \\ 55 & \quad 22 \\ \log &= .49714\,98726.88\end{aligned}$$

(correct value of last two figures = .94).

The labour saved by the small field of the Tables goes far to balance that occasioned by the additional steps in the calculation.

[502]

On the Surfaces divisible into Squares by their Curves of Curvature.

*[From the Proceedings of the London Mathematical Society, vol. IV. (1871–1873), pp. 8, 9.
Read December 14, 1871.]*

GEOMETRICALLY, the theorem may be stated as follows:—Consider any two curves of curvature AB, CD of one set, and any two AC, BD of the other set, as shown by the continuous lines of the figure: drawing the consecutive curves as shown by dotted lines, the curve consecutive to AB at an arbitrary (infinitesimal) distance from AB , the other three curves may be drawn at such distances that the elements at A, B , and C shall be each of them a square; but this being so, the element at D will not in general be a square, and it is only for certain surfaces that it is so. But if (whatever the curves

[Figure: net of curvature lines on a surface — diagram omitted]

of curvature AB, CD, AC, BC may be) the element at D is a square, then it is clear that the whole surface can be, by means of its curves of curvature, divided into infinitesimal squares.

Analytically, if for a given surface the equations of the curves of curvature are expressed in the form $h = f(x, y, z)$, $k = \phi(x, y, z)$, then the coordinates x, y, z can

be expressed each of them as a function of the parameters h, k , and we have for the element of distance between two consecutive points on the surface

$$ds^2 = dy^2 + dz^2 = A dh^2 + C dk^2,$$

where A, C are in general each of them a function of h and k . The condition for the divisibility into squares is that the quotient $A \div C$ shall be of the form function $h \div$ function k .

It was shown by M. Bertrand that, in a triple system of orthotomic isothermal surfaces, each surface possesses the property in question of divisibility into squares by means of its curves of curvature. But in such a triple system, each surface of the system is necessarily a quadric; so that the theorem comes to this, that a quadric surface is, by means of its curves of curvature, divisible into squares. The analytical verification is at once effected; taking the equation of the surface to be

$$\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} = 1,$$

then the expressions for the coordinates in terms of the parameters h, k of a curve of curvature are

$$\begin{aligned} x^2 &= \frac{a(a+h)(a+k)}{(a-b)(a-c)}, \\ y^2 &= \frac{b(b+h)(b+k)}{(b-a)(b-c)}, \\ z^2 &= \frac{c(c+h)(c+k)}{(c-a)(c-b)}, \end{aligned}$$

and we have

$$A(dx^2 + dy^2 + dz^2) = (h-k) \left[\frac{h dh^2}{(a+h)(b+h)(c+h)} + \frac{k dk^2}{(a+k)(b+k)(c+k)} \right],$$

so that $A \div C$ is of the required form.

But there is nothing to show that the property is reduced to quadric surfaces, and the question of the determination of the surfaces possessing the property appears to be one of considerable difficulty, and which has not hitherto been examined.

[503]

On the Surfaces each the Locus of the Vertex of a Cone which passes through m Given Points and touches $6 - m$ Given Lines.

*[From the Proceedings of the London Mathematical Society, vol. IV. (1871–1873), pp. 11–47.
Read January 11, 1872.]*

I consider the surfaces, each of them the locus of a vertex of a (quadri)cone which passes through m given points and touches $6 - m$ given lines; viz. calling the given points $a, b, c, \dots$, and the given lines $\alpha, \beta, \gamma, \dots$, there are the surfaces in question are:

	Order
$abcdef$,	4
$abcde\alpha$,	8
$abc\alpha\beta\gamma$,	16
$abc\alpha\beta\gamma$,	20
$ab\alpha\beta\gamma\delta$,	24
$a\alpha\beta\gamma\delta\epsilon$,	24
$\alpha\beta\gamma\delta\epsilon\zeta$,	16

I remark that the orders of these several surfaces are in effect determined by the investigations of M. Chasles in regard to the conics in space which satisfy seven conditions. The surface $abcdef$ may be considered as the “Mémoire on Quartic Surfaces,” [441]; and is the same Memoir as that of 4 “Mémoire on Quartic Surfaces,” [441]; and is the same Memoir as that of 4 in my “Mémoire on Quartic Surfaces,” but the other surfaces, and also the surfaces $a\alpha\beta\gamma$ and $a\alpha\beta\gamma\delta\epsilon$ may be I think considered as new. Are considered by Dr Hierholzer in his excellent paper “Ueber Kegelschnitte im Raume,” *Math. Annalen*, t. ii. (1870), pp. 568–586, and to him are due the equations given in the sequel for the surfaces $abdef$ and $abdef$; the researches of the present Memoir are in fact a continuation and development of those in the Memoir last referred to.

Table of Singularities, and Explanations in regard thereto.

1. We have on the below-mentioned surfaces respectively certain simple or multiple points, right lines, and curves, as shown in the following Table.

*[Table 1: complex tabular array of singularities for the seven surfaces — omitted, see scan
p. 100]*

2. In the Table, the upper margin refers to the surfaces, and the left-hand margin to the points, lines, and curves situate on these surfaces respectively; the body of the Table showing the number, and in () the multiplicity, of those points, lines, and curves in regard to the several surfaces respectively. Thus, points a ; for the surface $abcdef$, $6 \times (2)$, there are 6 such points, each of them a 2-conical (ordinary conical) point on the surface: so $abcde\alpha$, $5 \times (4)$, there are 5 such points, each a 4-conical point on the surface (viz. instead of the tangent plane there is a quartic cone); and so on. Similarly, lines ab (viz. these are the lines joining two points a, b); for the surface $abcdef$, $15 \times (1)$, there are 15 such lines, each a simple line on the surface; surface $abcde\alpha$, $10 \times (2)$, there are 10 such lines, each a double (ordinary nodal) line on the surface; and so on. We have in two places the multiplicity $(2 + 2)$, which refers to a tacnodal line, as presently explained. The corner letters C, P, L denote respectively proper cone, plane-pair, and line-pair, as afterwards explained.

3. The lines and curves referred to in the left-hand margin are:

- (1) ab , line joining the points a and b .
- (2) α , line α .
- (3) $[ab, \alpha, \beta, \gamma]$, pair of lines meeting each of the four lines, or say the tractors of the four lines $ab, \alpha, \beta, \gamma$. As regards the surface $aba\beta\gamma\delta$, the multiplicity is given as $(2 + 2)$, viz. the line is (not an ordinary nodal, but) a tacnodal line, each sheet touching along the whole line the hyperboloid $a\beta\gamma$.
- (4) $[\alpha, \beta, \gamma, \delta]$, tractors of the four lines $\alpha, \beta, \gamma, \delta$.
- (5) $[ab, cd, \alpha, \beta]$, tractors of the four lines ab, cd, α, β .
- (6) abc, def , line of intersection of the planes abc and def .
- (7) $abc, de\alpha$, line in the plane abc joining the intersections of this plane by the lines de and α respectively.
- (8) $abc, \alpha\beta$, line in the plane abc joining the intersections of this plane by the lines α and β respectively. As regards the surface $abc\alpha\beta\gamma$, the multiplicity is given as $(2 + 2)$, viz. each line is (not an ordinary nodal, but) a tacnodal line, each sheet touching along the whole line the plane abc .
- (9) Cubic $abcdef$, cubic curve through the six points a, b, c, d, e, f , common intersection of the cones each having its vertex at one of these points and passing through the others belonging to the lines abc .
- (10) Quadriquartic $\alpha\beta\gamma, \delta\epsilon\zeta$, intersection of the quadric surfaces $\alpha\beta\gamma$ and $\delta\epsilon\zeta$, that is, the quadric surfaces formed by the four lines $\alpha, \beta, \gamma, \delta$ and δ, ϵ, ζ respectively.
- (11) Excuboquartic $\alpha\beta\gamma, \delta\epsilon, a$, quartic curve generated as follows: viz. taking any line whatever which meets the lines α, β, γ , the point of the line meets α, β, γ in a point P , and a plane through a in the lines δ, ϵ in a point P ; the locus the points P in question (theory further considered in the sequel).

Special forms of Quadricones.

4. We have to consider the special forms of (quadri-)cones; these are: 1° The sharp-cone, or plane-pair; that is, a pair of two planes, intersecting in a line called the axis, the vertex being in this case an indeterminate point on the axis. Observe that a plane-pair passes through a point, or touches a line; therefore there are six conditions through such a point; it touches a given line through any line in either of the two planes through it. 2° The flat-cone, or line-pair; this is a pair either of two distinct lines (not meeting through a given point and touching 4 given lines), or, in the same case of two distinct lines), or, in the same case of two coincident lines), or two coincident lines, intersecting in a point called the vertex. And we have to consider in this connexion the 3° axis-pair; that is, a pair of planes intersecting in a line, called the axis, of the axis-pair; viz. the axis is a line meeting a given line.

Singular Lines and Curves on the Surfaces.

5. We may establish *θ priori* the existence, and even to some extent the multiplicity of the several lines and curves on the surfaces $abcdef \dots \alpha\beta\gamma\delta\epsilon$ Thus:

1. Lines ab : take for the vertex of the cone a point at pleasure on the line ab ; the cone passing through b will (per finite pass through a) and the conditions are then that the cone shall touch the line cd , α , β , the order 4; surface $abcde\alpha$, ditto $a\beta\gamma\delta$, surface $a\alpha\beta\gamma\delta\epsilon$ and $a\beta\gamma\delta\epsilon$ surface $abcde\alpha$, and so on for the remaining 10 surfaces; the number of the remaining tangent planes is $= a - 4$. Hence as regards the surface $abcde\alpha$, the cone touches a tangent planes is the line ab , by line α , β ; in the next case there is besides $a - 4 = 2$, this is besides one line through the point of the cone a , α ; in the next case there is besides one line in abc , $\alpha\beta$, γ , but these touch the lines α , β , γ , and the other line in the plane abc , $\alpha\beta$, γ , but these touch the lines α , β , γ .

2. 2° Lines $[ab, \alpha, \beta]$: taking the vertex in the line of these tractors, the cone meets a proper cone, but it is a plane-pair having for axis the tractor: the surface $abc\alpha\beta\gamma$ is a line, and that the surface $a\alpha\beta\gamma$ is a line, and that the surface $a\alpha\beta\gamma$ is a line, also on the surface.

Surface $abc\beta\gamma$. Cone is a plane-pair; the two planes intersecting in the line ab , and the two planes through the planes the points α, β, γ ; the other is through the point: a . The vertex being an indeterminate point on the line: there is an indeterminate point.

Surface $abc\beta\gamma$. Cone is a plane-pair; the two planes intersecting in the line ab , and passing the one of them through α and the other through β . Hence the line is a double line on the surface.

3. 3° Lines $[\alpha, \beta, \gamma, \delta]$: taking the vertex in one of these tractors, then as in the last case, the cone is either a line-pair having the tractor for its vertex, or else a plane-pair having for axis the tractor; viz. the three cases are *Surface $a\alpha\beta\gamma\delta$* . Cone is a line-pair, the one line being α, β , the other a line in the plane through the tractor and the point a , meeting the lines γ and δ .

Surface $\alpha\beta\gamma\delta\epsilon$. Cone is a line-pair, one line being α, β , the other a line in the plane through the tractor and meeting each of the lines γ and δ . In each case the line is an indeterminate point.

Surface $\alpha\beta\gamma\delta\epsilon\zeta$. Cone is a plane-pair, the two planes intersecting in the tractor, and passing through the planes the lines δ and ϵ respectively. Hence the line is a nodal line on the surface.

4. 4° Line abc, def . Cone is a plane-pair; consisting of the two planes abc and def .
5. 5° Line $abc, de\alpha$. There are two cases: *Surface $abcdef$* . Cone is a plane-pair, the two planes intersecting in the line; the one being abc , and the other passing through the points a . *Surface $abcde\alpha$* . Cone is a plane-pair, the two planes intersecting in the line; the one being abc and the other passing through de . In each case the line is a line on the surface simply.
6. 6° Cubic $abcdef$. The cone has its vertex on the cubic which is the locus of the vertices of cones each through the six points a, b, c, d, e, f . Hence the cubic is on the surface $abcdef$.
7. 7° Quadriquartic $\alpha\beta\gamma, \delta\epsilon, \zeta$. Cone is a line-pair, consisting of two lines through the point of the quadriquartic curve, the one meeting each of the lines α, β, γ , and the other of these meeting each of the lines δ, ϵ, ζ .
8. 8° Excuboquartic $\alpha\beta\gamma, \delta\epsilon, a$. Cone is a line-pair, consisting of two lines through the vertex on the excuboquartic, the one being the line meeting the three lines α, β, γ , and the other meeting the two lines δ, ϵ , and passing through the point a .

Mode of obtaining the several Equations: Notations and Formulæ.

6. The equations of the several surfaces are obtained by taking an centre of projection an assumed position of the vertex, and projecting everything upon an arbitrary plane; the projections of the given points and lines are points and lines in the arbitrary plane, and the section of the cone by this plane is a conic; the equation of the surface is then obtained as the condition that there shall be a conic passing through a given points and touching $6 - n$ given lines. We obtain at once, as the plane of projection any one of the coordinate planes; or for greater generality, the plane of projection $x = 0$, $y = 0$, and the centre of projection a point (x_0, y_0, z_0, w_0) . The point of the plane $z = 0$ corresponding to a point (x, y, z, w) of space is

$$(x_0z - xz_0, y_0z - yz_0, 0, w_0z - wz_0);$$

or, what is the same thing, taking $w = 1$, the projection of (x, y, z) is

$$(x_0 - x, y_0 - y, 0, w_0 - z),$$

which I shall represent by ξ, η, ζ .

7. We have similarly for the coordinates of the points (a, b, c, w) , say the points a :

$$P = bz - z_0aw,$$

$$Q = bz - x_0cw,$$

$$R = wz - z_0cw,$$

$$S = ay - bz,$$

($P_a = b_a y - y_a c$, etc.). We have similarly for the projection of the point on the plane $w = 0$, and touched by the conic.

The following notations and formulæ are convenient:

8. $pabc = 0$ is the equation of the plane through the points a, b, c ; viz.

$$pabc = \begin{vmatrix} z, & y, & z, & w \\ a_a, & b_a, & c_a, & w_a \\ a_b, & b_b, & c_b, & w_b \\ a_c, & b_c, & c_c, & w_c \end{vmatrix}$$

Of course $pbac = -pabc$, &c. Observe that here, and in the notations which follow, the letter p is used in referring to the coordinate (x, y, z, w) , and that the index of p (... when in a given is expressed) shows the degree in these coordinates.

10. $pza = 0$ is the equation of the plane through the point z and the line α ; viz.

$$pa = P_a x + Q_a y + R_a w = S_a w,$$

where

$$P_a = b_a y_a - c_a w_a,$$

$$Q_a = -a_a,$$

$$R_a = a_a y_a,$$

$$S_a = a_a c_a - b_a w_a,$$

$$S = a_a c_a + b_a c_a;$$

and (a, b, c, f, g, h) are the coordinates of the line α ; observe that $paa = pba$, where these (and (x, y, z, w) of any line are connected by an identity equation.

11. $p^4 ab = 0$ is the equation of the quadric surface through the points a, b and the lines α, β ; viz.

$$\begin{aligned} p^4 ab = & (xy - zy)P_a P_b \\ & - (xy - yz)Q_a Q_b \\ & - (yz - zx)R_a R_b \\ & + (zy - xz)S_a S_b \\ & - (a, b, f, g, h c)(a + b, e + f, g + h) \\ & + (e, b, f, g, c)\beta(\beta_p, \beta_q, \beta_r); \end{aligned}$$

where

$$abc = \begin{vmatrix} x_a, & y_a, & z_a \\ x_b, & y_b, & z_b \\ x_c, & y_c, & z_c \end{vmatrix}, \text{ etc.}$$

$(a_a, b_a, c_a, f_a, g_a, h_a)$ are, as before, the coordinates of the lines $\alpha, \beta, (a_a, (a_a), \dots)$ being the coordinates of the given line $p^4 ab$, &c.

12. It is to be noticed that, writing

$$\begin{aligned} P &= ay = pz + uw, \\ Q &= -bz + ay + tw, \\ R &= -ay + bx + sw, \\ S &= az - by + cz, \end{aligned}$$

viz. $P_a = b_a y + a c x$, &c., the line α being $a_a x + p_a y + r y + a w = 0$, $p_a y + r y = 0$, and further that we have identically

$$-pPaQ = L_a x P_a + M_a Q_a + N_a R_a + \Omega_a S_a,$$

where

$$\begin{aligned} L &= (g^2 + 2f^2 + a^2)x + (cf + bg + ah)w + (bc + af) \\ M &= -(cg + e)x + (g^2 + 2f^2 + a^2)y - (ef + g) \\ N &= -a(bx + e)x - (eg + h)y + (g^2 + 2f^2 + a^2)z - (ag + ah) \\ \Omega &= -(cf + bh)x - (ef + g)y - (ag + ah)z + (g^2 + 2f^2 + a^2) \end{aligned}$$

and L_a , c. are the values of L , c. on substituting therein $(a_a, \dots)$ and $(x_a, \dots, w_a)$ for the unaccented and accented letters respectively.

13. Observe that we have

$$pabc = (Lpx)(MR - LS) + (\dots) = -(QR - LR + RS - \dots) - SP = pbc;$$

and similarly

$$L_a x b_a + l_a y b_a + l_a z c_a + l_a w c_a + l_a u c_a + l_a v c_a + l_a u c_a + l_a v c_a + l_a w c_a = (a, c, e, f, g, h b)(g+i)(g+j)(g+l)(g+m)(g+n),$$

$$\begin{aligned}
 & -hM + gN - cb\Omega + (af + by + ch)p = +p_a, \\
 & -fL + eM - cP + (af + by + ch)p = +p_a, \\
 & -gL + fN - aS + (\dots)p = +p_a, \\
 & -cL - bM + aN + (\dots)p = +p_a.
 \end{aligned}$$

14. $p^4ab = 0$ is the equation of the cubic surface through the points a, b, c and lines α, β (cab the in the line) the points one of p is the equation of a curve b, c, α, β is at most $1 + 3 = 5$ points; observe also that the equation determines this cubic surface in a unique manner, taking the lines α, β in the order α, β ; if $1 + p$ points were given, then there are 3 other points on η , 3 points on ζ , and 3 lines satisfy in like manner the equation of the cubic, surface in question.

15. We have

$$p^4ab \cdot p^4cd = \begin{vmatrix} a, & b, & e, & e \\ a_a, & b_a, & c_a & \\ a_b, & b_b, & c_b & \\ a_c, & b_c, & c_c & \\ a_d, & b_d, & c_d & \end{vmatrix},$$

viz. this determinant, equated to zero, gives the equation of the surface.

To prove this, take as before the unaccented letters (a, b, c, f, g, h) to refer to the lines α , and the letters with one, two, and three accents to refer to the lines β, γ, δ respectively. Referring to the foregoing expressions for L, M, N, Ω , and observing that for a point on $L = a^4 = b^2 + c^2 + a^2 = b^2, c.$, the values for a point on the line β are: $L = a^4b^2c^2, \Omega = a^2a^2$. Whence at β to substitute thus exactly the values of L, M, N, Ω respectively; the values are $a^2b^2c^2, \Omega = a^2a^2$. And, in the other manner, $L^2 = c^4, D^4 = c^4, L_a = L_a, M_a = M_a$, and three values satisfy the equation of L, M, N, Ω . In a surface passing through the lines β, γ, δ . Moreover, the surface contains the lines α and the line β is also a triple line.

To show that the surface passes through the line α and that the coefficients of the points α be in b, c, d , the equation reduces itself to a single one in the variables L, M, N, Ω, c . We have, in fact, $a_ax_a + a_bx_b + a_cx_c + a_dx_d - b_ay_a - b_by_b - b_cy_c - b_dy_d = (LP)(LP)(LP)$. And the equation of the surface, writing $a_ax_a + b_ay_a + c_ax_a + d_ay_a = 0$, becomes

$$\begin{vmatrix} a, & y, & e, & e \\ L, & M, & N, & \Omega \\ L_a, & M_a, & N_a, & \Omega_a \end{vmatrix} = 0.$$

$$\begin{aligned}
 & -LM + g'N + e''\Omega + (a'f + b'y + ch')p = +p_a, \\
 & -fL + eM - cN + (af + by + ch')p = +p_a, \\
 & -gL + fM - cN + (cf + by + ch')p = +p_a, \\
 & -cL - bM + aN + (\dots)p = +p_a.
 \end{aligned}$$

14. $p^4abc = 0$ is the equation of the cubic surface through the points a, b and the lines α, β, γ ; the line b is here a 5 line, $b, 4, 9$ points: C may be at most 9, but α, β are 5 points respectively; 3 other points on η , 3 on ζ , and 4 on a surface; in like manner, 3 other points on η , 4 on ζ , and 3 on a ζ, η . Observe that the equation determines this cubic in a unique manner.

15. We have

$$pza \cdot p^2\beta\gamma = \begin{vmatrix} x, & y, & z, & w \\ x_a, & y_a, & z_a, & w_a \\ x_\beta, & y_\beta, & z_\beta, & w_\beta \\ x_\gamma, & y_\gamma, & z_\gamma, & w_\gamma \end{vmatrix},$$

$$\therefore pa \cdot p^4\beta\gamma + p^4\beta \cdot p^2a + p^4\gamma \cdot p^2a + p^4a \cdot p^2\beta\gamma = 0,$$

of the six letters (say z, y, z, w, z', z''); but the equation may be written

$$pza \cdot p^2\beta\gamma + p^4\beta \cdot p^2a\gamma + p^4a \cdot p^2\beta\gamma = 0,$$

which is of the order 4 in regard to (x, y, z, w) .

16. To avoid confusion with the geometrical term *line*, I will speak (not of the lines, but) of the *rows* of these tables; thus the multiplicity of contains only 1, 2 is in general p a -2 , but it is on the second row that the multiplicity changes, and only a row containing only 1, 2 is given by p Zeuthen, but which I have completed so far as regards the double lines as follows:

[Table 2: "Plane Section of the Torse" — 9-row tabular listing, see scan p. 82]

17. To avoid confusion with the geometrical term *line*, I will speak (not of the lines, but) of the *rows* of these tables; thus the multiplicity of contains the foregoing Plückerian relations. To fix the steps, I propose to discuss in the sequel both these Plückerian relations. To fix the ideas, I shall consider it as made, so that the osculating planes will pass through this line, in respect of the curve a stationary tangent in respect of the torse, c .

20. Next, for the column 2, we have besides $r - 2$ stationary points; this is, in fact, plain; the number of tangent planes which pass through a fixed point in the plane is still $= n$. For Class 3, 7, 8, the plane is a stationary planes; viz. a tangent among the tangent planes which pass through a fixed point thereof, and the number of the remaining tangent planes is $n - 3$. And so for Classes 4 and 5, the plane contains for n among the tangent planes which pass through the fixed point thereof, and the number of the remaining tangent planes is $n = n - 2$.

21. Next as to the r and a columns.

Case (2). The plane passes through a generating line, and therefore besides cuts the torse in a curve of the order $r - 1$; this generating line is a stationary tangent line of the torse, viz. counts r times among the r tangents (in regard to the curve $r - 1$) in the plane. The remaining tangents (in regard to the curve $r - 1$) are not in general osculating ones of any of the planes; their number is r and there are besides $\beta + 1$ stationary tangents. In Case (4) the plane cuts the torse in the generating line counting twice, and in a curve of the order $r - 2$; the generating line is a stationary tangent line of the torse, c.

Case (3). The plane cuts the torse in the generating line counting twice, and in a curve of the order $r - 2$; the generating line is a nodal curve on the torse, an ordinary tangent at the point of contact with the curve m (so that the number of stationary tangents remains $= \alpha$); and besides cuts the curve in $r - 4$ points as in Case 2.

Case (4). The plane cuts the torse in two generating lines, each counting twice, and in a curve of the order $r - 4$; each of the generating lines is in regard to this curve an ordinary tangent at the point of contact with the curve; in the remaining points the section consists of an ordinary tangent at the point.

Case (5). The plane cuts the torse in a generating line counting 3 times, and in a curve of the order $r - 3$; the generating line is a stationary tangent in regard to this curve.

Case (6). The plane meets the torse in a generating line counting twice, and in a curve of the order $r - 2$; the generating line is in regard to this curve a tangent at the point of contact with the curve; counting 4 times, and besides meeting it in $r - 6$ points.

Case (7). The plane meets the torse in the generating line counting 3 times, and in a curve of the order $r - 3$; the generating line is in regard to this curve a stationary tangent at the point of contact with the curve, counting 3 times, and besides meeting it in $r - 6$ points. The whole number of stationary tangents is thus $= a + 1$.

Case (8). The plane meets the torse in a generating line counting twice and in a curve of the order $r - 2$; the generating line is in respect of this curve a tangent counting 3 times, the points of contact with the curve m , each of these is besides $r - 5$ intersections. Moreover, the generating line counting as 2 stationary tangents, the lines is to be $= a + 2$.

Case (9). The plane meets the torse in a generating line counting 3 times, and in a curve of the order $r - 3$. The generating line is in respect of the curve an ordinary tangent at one of the points of contact with the curve m , viz. the point at which the plane is a tangent plane of the torse; this is, $S = b$ and S is which sections, and it is also a stationary tangent at the other of the points of contact which the plane is a stationary tangent of the curve m , viz. at which point the plane meets a in fact a stationary tangent at the other of the points of contact with the curve m (viz. the point at which the plane is not a tangent plane of the torse).

22. In the foregoing the discussion of Cases (1) and (2) is satisfactory; but in the other cases there is some difficulty which it has not been worth my while to examine. The Salmon-Cremona equations as I have stated them are

$$\begin{aligned}a(r-2) &= 2a + 4\beta + g + 8a + 4H, \\ \sigma(r-2) &= 4n - 4\beta + g + 8a + 14H, \\ &\text{etc.}\end{aligned}$$

which can be shown to hold; thus considering the equation $z(r-2) = 4n - 4\beta + g + 8a + 14H$, the stationary point only counts here as 4, the stationary point only counts as 4 in regard to the order of the cone, so that the number of the stationary planes is in this case $= 4\beta$; and the case is $= 4\beta - 4\beta + g + 8a + 14H = 4n - 4\beta - 8a + 14H$.

25. We have similarly the following Table:

[Table 3: “Cone through the Curve” — tabular listing, see scan p. 81]

26. The vertex of the cone m at m being any point not on the curve, the cone has for each of the points (1) of the curve, viz. besides the apparent double point and h double points; for each of the points σ a stationary line; for each of the points ζ a 3-pointic intersection with the line a ; for each of the points ζ , a 3-pointic intersection where, as before, $(a, a, \zeta, \eta, \zeta)$ etc., respectively are referred (from the cone) by aid of the foregoing Plückerian relations.

Geometrical Theory of the foregoing Relations (continued).

27. We have

$$\begin{aligned} y' &= n - 8(n - r) - 4(g - H), \\ g' &= y' + 6(m - r), \\ y'(r - 1) &= y - (n - r)(n + 2r - 5) + 2(g - H), \\ y &= -\frac{1}{4}(m - n)(m - r - r) + (g - r)(g - r) + (g - r) - 2(g - H). \end{aligned}$$

28. Instead of obtaining the above values of y, β, g directly it is convenient to verify them by substitution in the equations from which they were obtained; viz. writing for shortness $a + 8(n - r)$ for b , these may be written

$$\begin{aligned} (n - r - 2) + 8a + 4H + 8a - 6x + 4H &= 0, \\ -2a(r - 2) - 8r + 4y - 4 - 2(M)r + 2 &= 0, \\ -2y(r - 2) + 8r - 4y - 11(8 - 14)H + (5r - 9 - 24H) &= 0, \\ -2y(r - 2) - 4y - 11(8 - 14)H + (5r - 9 - 24H) &= 0, \\ -8r + 16r + 18a(r - 6) - (3r - 4(8 - 18r - 12r + 9r - 25 + 24H) &= 0, \\ -4y(r - 2) - 4y - (r - 6)(r - 4) + 4y(8 - 14)H + 12H + 24H &= 0. \end{aligned}$$

29. At a point H I consider any section through the arbitrary point and z ; in effect any section through a , the line a being a 3-pointic line, a has a 3-pointic intersection (whence is a stationary tangent, and so a curve $a + 2$ on the surface); but it is to be the case is, the line from the arbitrary point to b has a 3-pointic intersection with the torse.

30. At each of the a points the tangent line with the curve $a + 2$ has so simply the cone a .

31. At points of the tangent line H I consider any section through the arbitrary point and z ; in effect any section through a , the line a being a 3-pointic line, a has a 3-pointic intersection (whence is a stationary tangent on the surface); but it is to be the case, the line from the arbitrary point to b has a 3-pointic intersection with the torse.

For a point H I consider any section through the arbitrary point and x ; in effect any section through a ; there is at a triple point and the line through a has then a 3-pointic intersection at a . Hence a line at a to a through a , viz. $= 2.4\beta$ such cuspidal lines through a , the corresponding lines of cusps in question. We have for the number of stationary lines $= 2g$.

32. But it is to be further shown that the curve m is torsoidal, each sheet of the surface being touched along the line by the plane $Z = 0$; we have to show that the

which, of course, satisfied by $\theta = 0$. Moreover the derived equation

$$= a - 2D\delta = \frac{Da^2}{d(b+d)^2a^2c'^2\left(\frac{a^2}{d}\right)^2} + \frac{aa^2c^2}{d(b+d)^2} + \frac{aa'a^2}{d(b+d)^2}$$

is also satisfied by $\theta = 0$, so that this is a double root. The equation in fact is

$$(Pa^2 + Pa - Da)(a^2 + a^2c'^2(B+a)(d+b)(d+c)) \\ + (Pa + db'^2(B+c)(d+a)(d+c) - 3a'^2c^2(B+a)(B+b)(d+c)) = 0,$$

or, expanding and dividing by θ^2 , this is

$$(Pa^2 + a)(d+a)(d+b)(d+c) + 2a'b'(d+a)(d+b)(d+c)c \\ - (Pb'^2 + ac'(b+c)c)(a' + c'(d+a)(d+b)(d+c)) = 0.$$

33. For a point H I consider any section through the arbitrary point and x ; in effect any section through a ; there is at a triple point and the line through a has then a 3-pointic intersection at a . Hence a line at a to a through a , viz. $= 2.4\beta$ such cuspidal lines through a , the corresponding lines of cusps in question. We have for the number of stationary lines $= 2g$.